

Tonal Languages

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Modeling

Probit Model

```
glm_tone_probit <- glm(data = tones_complex,
                      formula = complex_tonal ~ dist_equator,
                      family = binomial(link = probit))
summary(glm_tone_probit)

##
## Call:
## glm(formula = complex_tonal ~ dist_equator, family = binomial(link = probit),
##      data = tones_complex)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -0.7904  -0.6925  -0.5813  -0.3552   2.1410
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.618014   0.103299  -5.983 2.19e-09 ***
## dist_equator -0.018829   0.004648  -4.051 5.10e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 475.43  on 526  degrees of freedom
## Residual deviance: 458.18  on 525  degrees of freedom
## AIC: 462.18
##
## Number of Fisher Scoring iterations: 6
```

Logit Model

```
glm_tone_logit <- glm(data = tones_complex,
                     formula = complex_tonal ~ dist_equator,
                     family = binomial(link = logit))
summary(glm_tone_logit)

##
## Call:
```

```

## glm(formula = complex_tonal ~ dist_equator, family = binomial(link = logit),
##      data = tones_complex)
##
## Deviance Residuals:
##      Min        1Q      Median        3Q        Max
## -0.7841  -0.6870  -0.5807  -0.3786   2.1221
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.02196    0.18206  -5.613 1.99e-08 ***
## dist_equator -0.03203    0.00878  -3.648 0.000264 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 475.43  on 526  degrees of freedom
## Residual deviance: 459.42  on 525  degrees of freedom
## AIC: 463.42
##
## Number of Fisher Scoring iterations: 5

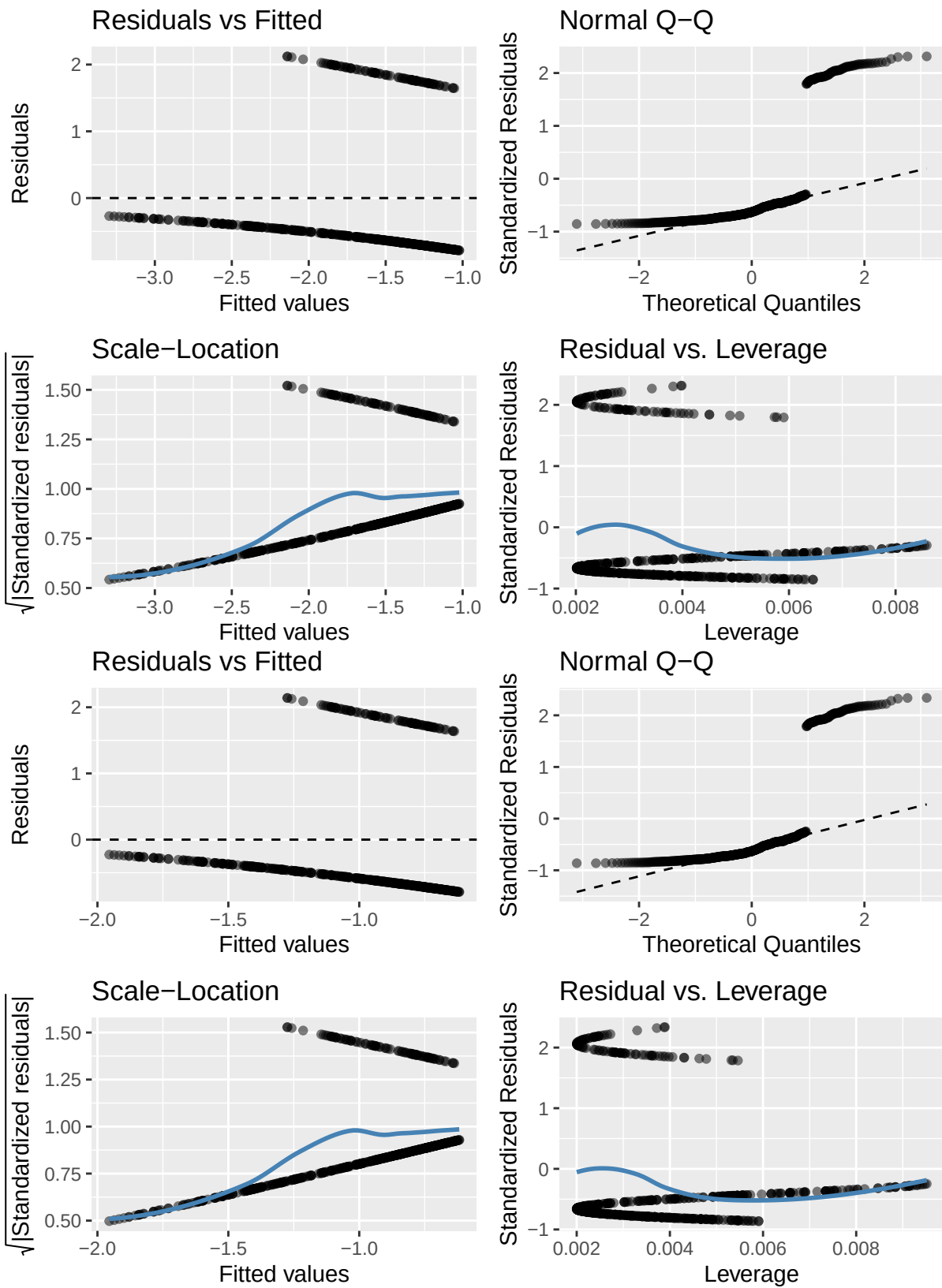
```

Transformations

```
glm_squared <- glm(data = tones_complex,
                   formula = complex_tonal ~ I(dist_equator^2),
                   family = binomial(link = probit))
summary(glm_squared)

##
## Call:
## glm(formula = complex_tonal ~ I(dist_equator^2), family = binomial(link = probit),
##      data = tones_complex)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -0.7424  -0.7169  -0.6202  -0.2209   2.1963
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.7034691   0.0811720  -8.666  < 2e-16 ***
## I(dist_equator^2) -0.0005245   0.0001208  -4.343  1.4e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 475.43  on 526  degrees of freedom
## Residual deviance: 446.95  on 525  degrees of freedom
## AIC: 450.95
##
## Number of Fisher Scoring iterations: 7
```

Residuals



Dispersion Estimation

Untransformed

```
# Estimate binomial GLM dispersion
qglm_linear <- glm(data = tones_complex,
                  formula = complex_tonal ~ I(dist_equator),
                  family = quasibinomial(link = probit))
summary(qglm_linear)

##
## Call:
## glm(formula = complex_tonal ~ I(dist_equator), family = quasibinomial(link = probit),
##      data = tones_complex)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -0.7904  -0.6925  -0.5813  -0.3552   2.1410
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.618014   0.100567  -6.145 1.58e-09 ***
## I(dist_equator) -0.018829   0.004525  -4.161 3.70e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasibinomial family taken to be 0.9477939)
##
##      Null deviance: 475.43  on 526  degrees of freedom
## Residual deviance: 458.18  on 525  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 6
```

Transformed

```
# Estimate binomial GLM dispersion
qglm_squared <- glm(data = tones_complex,
                  formula = complex_tonal ~ I(dist_equator^2),
                  family = quasibinomial(link = probit))
summary(qglm_squared)

##
## Call:
## glm(formula = complex_tonal ~ I(dist_equator^2), family = quasibinomial(link = probit),
##      data = tones_complex)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -0.7424  -0.7169  -0.6202  -0.2209   2.1963
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.7034691  0.0767864  -9.161 < 2e-16 ***
```

```

## I(dist_equator^2) -0.0005245  0.0001142  -4.591 5.52e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasibinomial family taken to be 0.8948624)
##
##      Null deviance: 475.43  on 526  degrees of freedom
## Residual deviance: 446.95  on 525  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 7

```